

COURSE NAME : DATABASE PRINCIPLES

COURSE CODE : CCS3400

CREDIT : 3(2+1)

**TOTAL STUDENT
LEARNING HOURS** : 120

PRE-REQUISITE : CCS3101

**LEARNING
OUTCOME** : Pelajar dapat :

1. Propose a database design using a selected data model. (C5)
2. Manipulate data in a database using Structured Query Language (SQL). (P4, DS)
3. Develop programs using stored procedures and triggers. (CTPS)

SYNOPSIS : This course covers database design using a selected data model and mapping to a relational database. It includes query formulation using SQL and programming with stored procedures and triggers.

COURSE CONTENT	<u>Total Lecture Hours:</u>
: 1. Database Design	2
- Definition and advantages of database approach	
- Database system environment	
- Database configuration	
2. Database Design	4
- Database design phases	
- Entity-Relationship (ER) model	
- ER diagrams	
- Enhanced Entity-Relationship (EER) model	
- Specialization	
- Generalization	
3. Relational Model	4
- Concept of relational model	
- Relational model constraints and database schema	
- Relational database design using ER	
- Relational database design using EER	
- Mapping to relational database	
4. Normalization	4
- 1NF	
- 2NF	
- 3NF	
- BCNF	
- Multivalued dependencies	
- 4NF	

5. Data Definition Language (DDL) and Data Control Language (DCL)	2
- Create tables with constraints	
- Statements to modify and drop tables	
- User access management	
6. Data Manipulation Language (DML)	2
- Statements to insert and update data	
- Statements to delete data	
- Statements to select data	
7. Data Other Objects in Databases	2
- View management	
- Sequence management	
- Index management	
8. Stored Procedure	4
- Uses of stored procedures	
- Stored procedure syntax	
- Using cursors	
- Stored procedure management	
- Execution of stored procedures	
9. Trigger Management	2
- Parts of SQL trigger	
- Event triggering	
- Trigger constraints	
10. Non-Relational Databases (NoSQL)	2
- Types and categories of NoSQL databases	
- SQL constraints and NoSQL advantages	
- Applications	
Total	28

**Face-to-face
learning Hour**

LAB	:	1. Database management system environment	3
		2. <i>DCL and DDL commands: create, drop, alter</i>	3
		3. <i>DML commands: insert, update, delete</i>	3
		4. <i>DML select command</i>	6
		5. <i>Subquery</i>	6
		6. Other schema objects	3
		7. Stored procedures	6
		8. Trigger	6

9. Non-relational databases 6

Total 42

ASSESSMENT : Course Work 70%
Final Exam 30%

REFERENCES : 1. Connolly, T. & Begg, C. (2021). *Database systems: A practical approach to design, implementation, and management* (6th ed.). Pearson-Prentice Hall.

2. Coronel, C. & Morris, S. (2022). *Database systems: Design, implementation, & management*. Cengage Learning, Inc.

3. Elmasri, R. & Navathe S. B. (2021). *Fundamentals of database systems* (8th ed.). Pearson-Addison Wesley.

4. Hoffer, J., Ramesh, V. & Heikki Topi. (2021). *Modern database management* (13th ed.). Prentice Hall.

5. Kroenke, D. & Auer, D. (2021). *Database processing: Fundamentals, design, and implementation* (16th ed.). Prentice Hall.

